# Yuxuan Liu

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## Education

**Shanghai Jiao Tong University (SJTU)** Sept. 2024 – June 2028

*B.S. in Computer Science and Technology, Zhiyuan Honors Program (John Hopcroft Class)*

- Honors: Zhiyuan Honors Program (Top 10% of the university); Zhiyuan Honors Scholarship (2024, 2025); Academic Excellence Scholarship (2025).
- Selected Coursework: Programming Language Design and Implementation (4.3/4.3), Machine Learning (4.3/4.3), Natural Language Processing and Large Language Model (4.0/4.3).

## Research

**Wave Forcing: Towards Speed-of-Light Streaming Video Generation**

Yijie Jin*, **Yuxuan Liu***, Jiajun Xu*, Ye Huang*, Chuzhi Wang, Jiayang Li, Ruiyang Chen, Chenling Hong, Xu Wang, Zhijie Deng.

*Preprint, 2026* *Equal contribution

- Wave Forcing pipelines chunk denoising and KV-cache refresh across GPU replicas; its WaveRT runtime overlaps KV communication with computation and balances VAE decoding across pipeline stages.
- WaveRT achieves **147.6 DiT FPS** and **134.8 end-to-end FPS** on 8 H200 GPUs with 4-step denoising and SageAttention: **7.07**× the throughput of the 4-step single-GPU baseline.
- Owned all model training and data selection, introduced the off-policy DMD stage and designed the mixing strategy. Implemented training infrastructure and participated in inference infrastrucutre optimization.

**Multi-Block Diffusion Language Models**

Yijie Jin, Jiajun Xu, **Yuxuan Liu**, Chenkai Xu, Yi Tu, Jiajun Li, Dandan Tu, Xiaohui Yan, Kai Yu, Pengfei Liu, Zhijie Deng.

*EMNLP 2026 Main Conference* (**Oral**)

- Combined MultiTF post-training with concurrent multi-block decoding, increasing LLaDA2-Mini's average tokens per forward pass from 3.47 to 6.19 and average accuracy from 79.95% to 81.03%.
- With Diffulex, increased average decoding throughput from 517.16 to 745.92 tokens/s (**1.44**×) across four math and code benchmarks on two H100 GPUs (TP=2, single-request decoding).

## Projects

**Diffulex: Diffusion Language Model Serving Engine** Apr. 2026 – Present

- Co-developed a diffusion LLM serving engine with block-level scheduling and paged KV caching, supporting concurrent refinement of multiple blocks within each request.
- The runtime uses a fixed-size Block Buffer to preserve static execution shapes for CUDA Graph replay while blocks are activated, refined, and committed to the KV cache.
- Integrated Fast-dLLM-v2 through pluggable decoding strategies, extending the engine with sub-block refinement and dual-cache execution.

**macchiatoBot: Multi-Agent Runtime** Nov. 2025 – Present

- Built an OS-inspired multi-agent runtime, separating agent reasoning from kernel-managed tool execution, permissions, and lifecycle control. Enabled per-session switching between local and remote workspaces via lightweight workers, preserving conversation context while redirecting shell and file operations.
- Served **100+ users** on SJTU's Shuiyuan Forum through concurrent agent instances, isolating per-session conversation contexts and user-scoped persistent memory and workspaces.

## Achievements

**4th Place**, SJTU HPC Comprehensive Ability Competition "HelloHPC" 2025

## Services

**Teaching Assistant**, Programming Language and Data Structure-III July 2026

**Head of Music Department**, SJTU Art Center July 2025 – June 2026